IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Executive Summary

- This project analyzes SpaceX Falcon 9 launch data to predict landing success.
- Data was collected using API and web scraping, then cleaned and prepared.
- Exploratory data analysis and SQL queries were used to identify patterns.
- Machine learning models were used to predict landing outcomes.
- The Decision Tree model achieved the best performance with ~94% accuracy.
- Results show that launch site, payload mass and booster version impact success.

Introduction

This project follows a full data science workflow including data collection, wrangling, analysis and machine learning.

- Objective: Predict Falcon 9 first stage landing success
- Importance: Helps reduce cost of space missions
- Dataset: SpaceX launch data
- Tools used:

  Python (Jupyter Notebook), SQL, Visualization tools

Section 1

Methodology

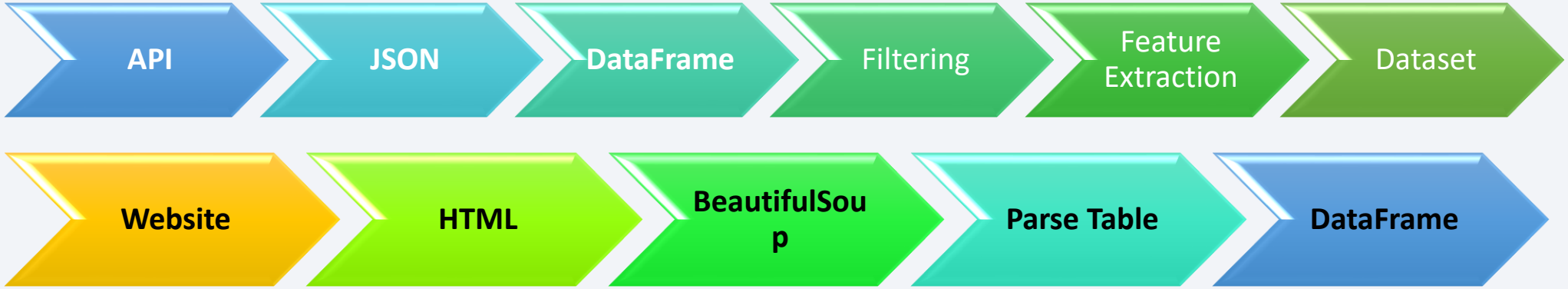
Methodology

Executive Summary

- Data collection methodology:
  - Collected SpaceX launch data using REST API and web scraping
- Perform data wrangling
  - Filtered Falcon 9 launches, cleaned the dataset, handled missing values and converted IDs into meaningful data
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Built models using training data
  - Tuned hyperparameters using GridSearchCV
  - Evaluated performance using accuracy and confusion matrix

Data Collection

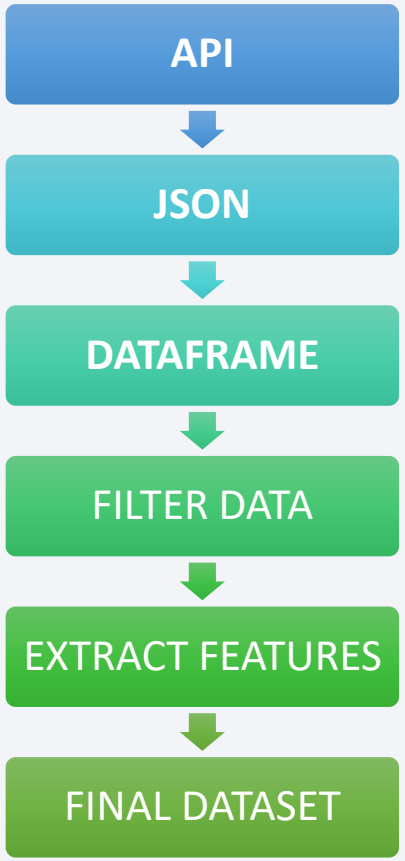
- Data collected using SpaceX REST API
- Used Python Requests library (GET request)
- Data received in JSON format and converted to DataFrame using pd.json_normalize
- Additional data collected from Wikipedia using web scraping (BeautifulSoup)
- Combined datasets used for further analysis



Data Collection – SpaceX API

- Retrieved and parsed SpaceX API data (JSON → DataFrame)
- Extracted additional features using API (Booster, Payload, Launch Site, Core data)
- Filtered records (single core & payload, valid date range)
- Filtered single core & payload launches, valid dates
- Kept only Falcon 9 launches
- Handled missing values (PayloadMass → mean)
- Prepared dataset for analysis
- **GitHub URL**

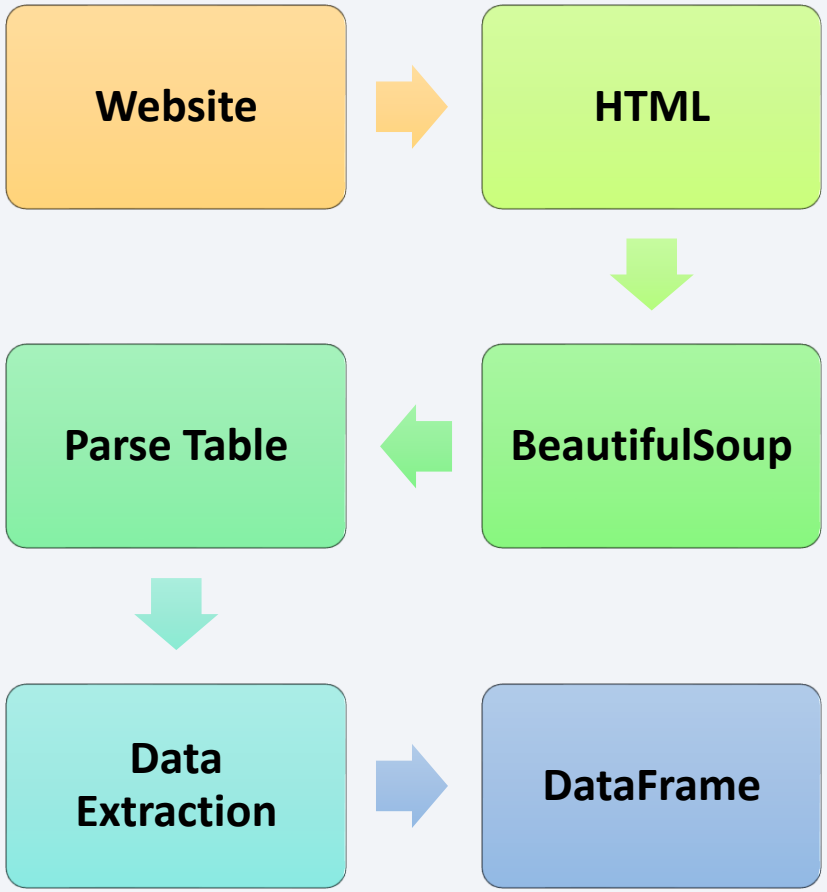
https://github.com/DivyaVKS/Coursera/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Collected additional launch data from Wikipedia
- Retrieved webpage content in HTML format
- Parsed HTML using BeautifulSoup
- Extracted launch table and column names
- Parsed table rows to collect launch records
- Cleaned and structured extracted data
- Converted data into Pandas DataFrame
- Exported dataset for further analysis
- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

- Loaded dataset and explored structure using Pandas
- Checked missing values → LandingPad (~29% missing), others complete
- Identified data types (numerical vs categorical features)
- Analyzed launches by site, orbit, and outcome
- Created binary label (Class: 1 = success, 0 = failure)
- Converted complex Outcome values into binary labels for ML
- Calculated overall success rate (~66%)
- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- Analyzed relationship between variables and landing success
- Flight Number vs Payload Mass → success improves with experience
- Flight Number vs Launch Site → performance varies by site
- Payload Mass vs Launch Site → payload distribution across sites
- Orbit vs Success Rate (Bar Chart) → identify high-performing orbits
- Payload Mass vs Orbit → impact of payload across orbit types
- Year vs Success Rate (Line Chart) → increasing success trend over time
- Tools used: Matplotlib and Seaborn for visualization, Pandas for data analysis
- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/jupyter-labs-edadataviz.ipynb

EDA with SQL

- Used SELECT, DISTINCT to identify unique launch sites
- Used WHERE, LIKE to filter specific launch records
- Used SUM and AVG to analyze total and average payload mass
- Used MIN(Date) to find first successful ground pad landing
- Used GROUP BY and COUNT to compare landing outcomes
- Used subquery with MAX to find boosters carrying highest payload
- Used substr() to extract month and year from dates
- Ranked landing outcomes using COUNT, GROUP BY, ORDER BY
- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Added circles and markers to show launch site locations
- Used green/red markers to show successful and failed launches
- Used MarkerCluster to manage overlapping launch records
- Added MousePosition to identify nearby feature coordinates
- Drew PolyLines to coastline, highway, railway and city
- Calculated distances to nearby features to study how location affects launch site choice
- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Added a dropdown menu to select launch site
- Added a range slider to filter payload mass
- Created a pie chart to compare success vs failure counts by launch site
- Created a scatter plot to show payload vs launch success
- Used Dash callbacks to update charts dynamically based on user input

WHY THESE WERE ADDED:

- To let users interactively explore launch performance by site, payload range and booster version.
- GitHub URL

https://github.com/DivyaVKS/Coursera/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

- Defined target variable (Class) and feature dataset
- Standardized input data using StandardScaler
- Split data into training and test sets
- Trained four models: Logistic Regression, SVM, Decision Tree, KNN
- Used GridSearchCV to tune hyperparameters
- Evaluated models using accuracy and confusion matrix
- Compared models and selected the best-performing model based on test accuracy

Data Preparation→ Feature Scaling → Train/Test Split → Model Training → Hyperparameter Tuning → Evaluation → Model Selection

- **GitHub URL**

https://github.com/DivyaVKS/Coursera/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

EXPLORATORY DATA ANALYSIS

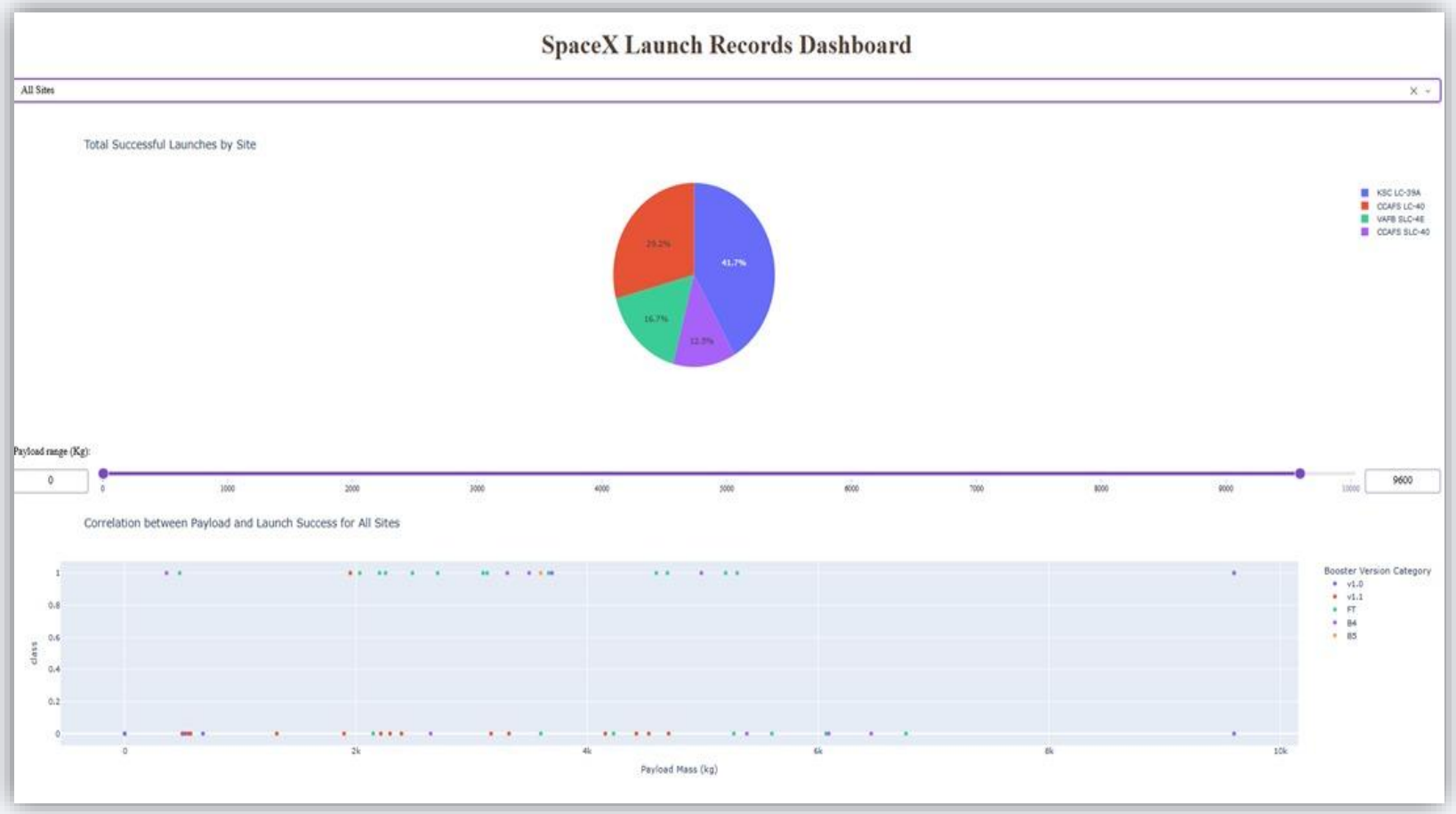
- Identified patterns between launch success and factors like flight number, payload mass, orbit and launch site.

INTERACTIVE ANALYTICS

- Dashboard and map visualizations show success trends, payload impact and geographical patterns of launch sites.

PREDICTIVE ANALYSIS

- Decision Tree achieved the highest accuracy (~94%).



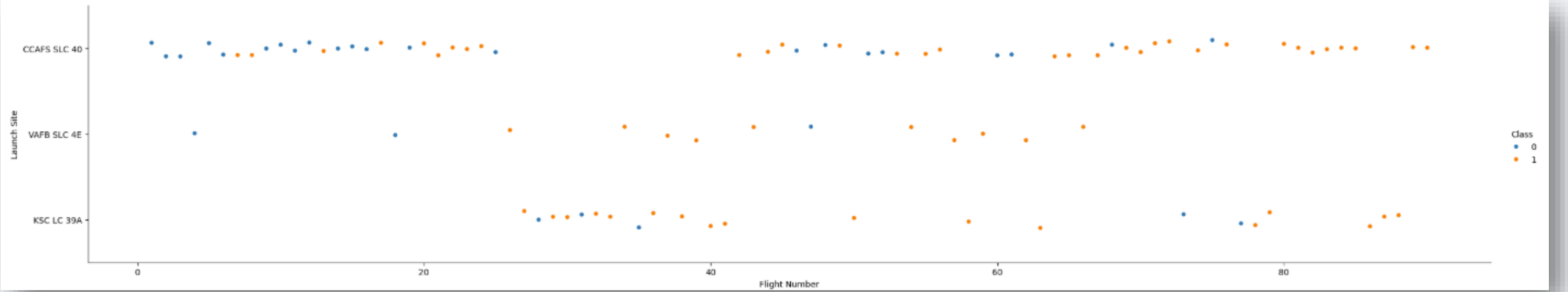
Interactive dashboard showing launch success patterns across sites and payload ranges

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

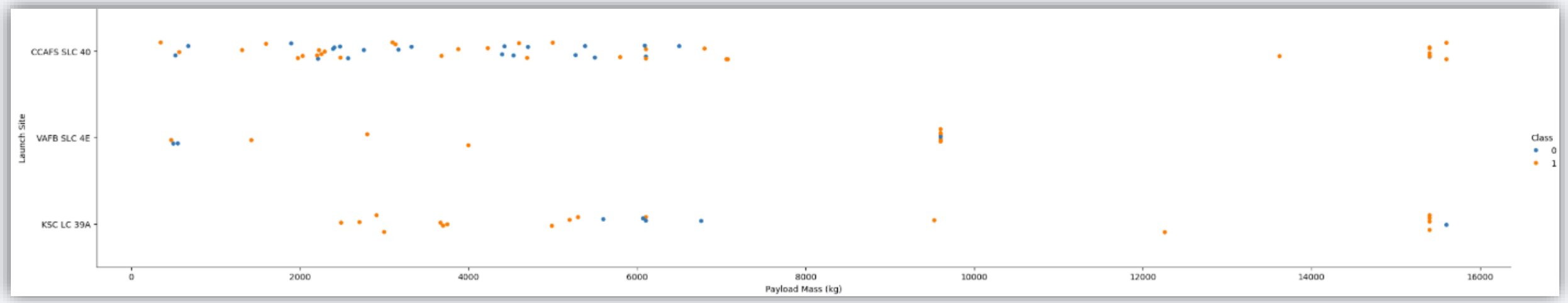
- The scatter plot shows that as the flight number increases; success rate improves across most launch sites.
- This means that with more launch experience, SpaceX achieved better landing outcomes.



Flight Number vs. Launch Site scatter plot

Payload vs. Launch Site

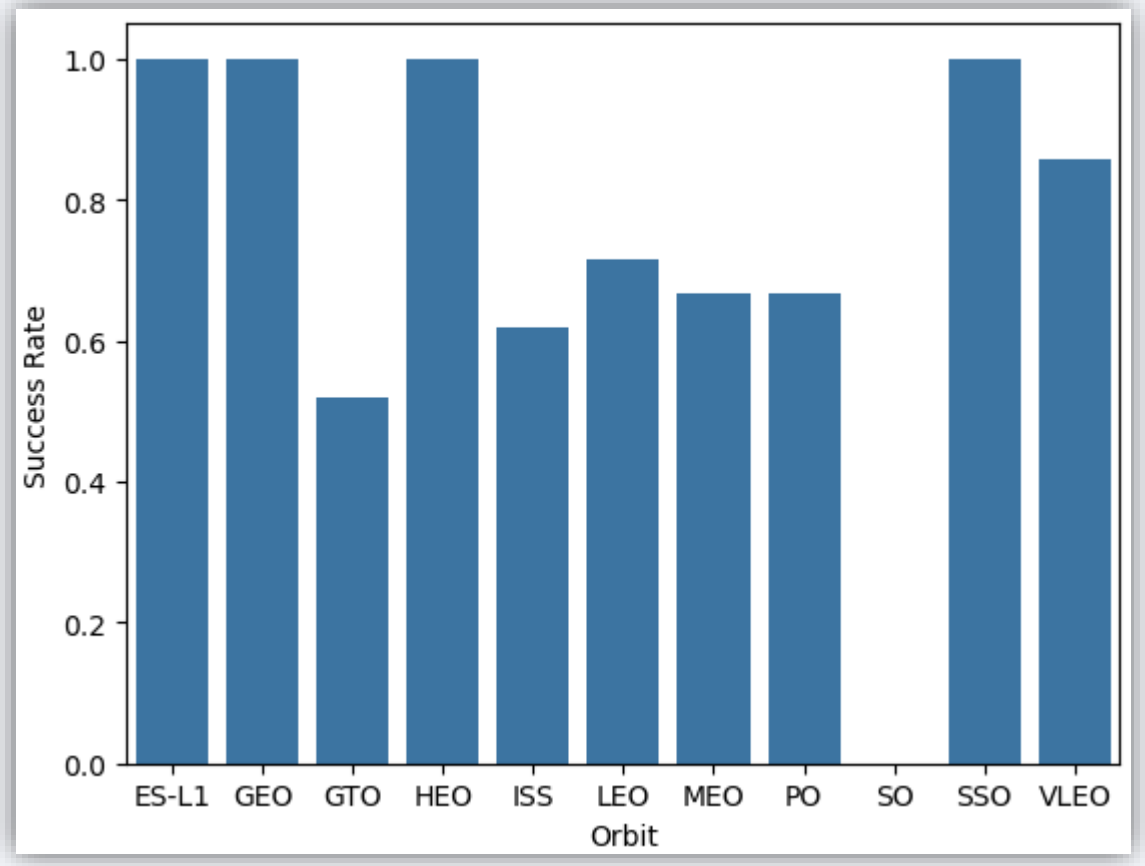
- The plot shows that payload mass varies across launch sites.
- VAFB-SLC launch site does not have launches with heavy payloads greater than 10000 kg.
- This suggests differences in mission profiles and launch configurations between sites.



Payload vs. Launch Site scatter plot

Success Rate vs. Orbit Type

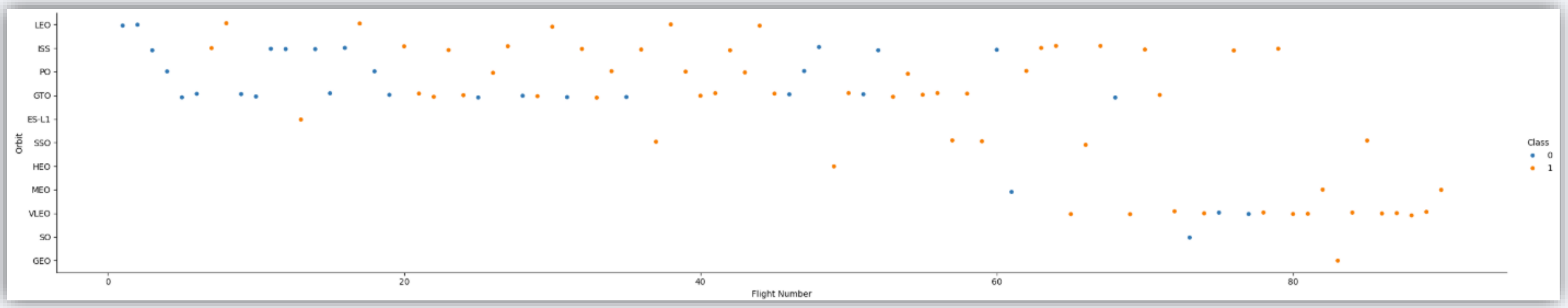
- Success rates vary across different orbit types
- Orbits such as ES-L1, GEO, HEO and SSO show the highest success rates
- Other orbits like GTO has low success rates



Success Rate vs. Orbit Type Bar Chart

Flight Number vs. Orbit Type

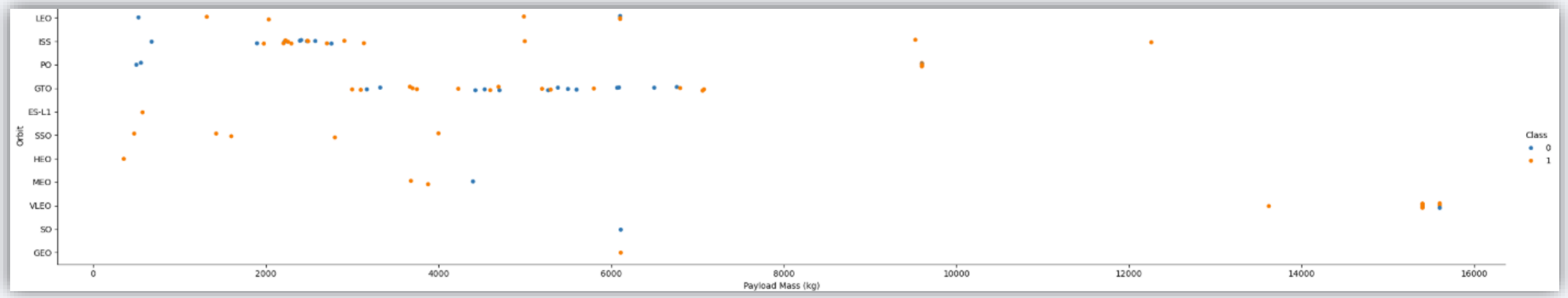
- The scatter plot shows the relationship between flight number and orbit type
- In LEO orbit, success increases with flight number (improvement over time)
- In GTO orbit, there is no clear pattern between flight number and success



Flight Number vs. Orbit Type scatter plot

Payload vs. Orbit Type

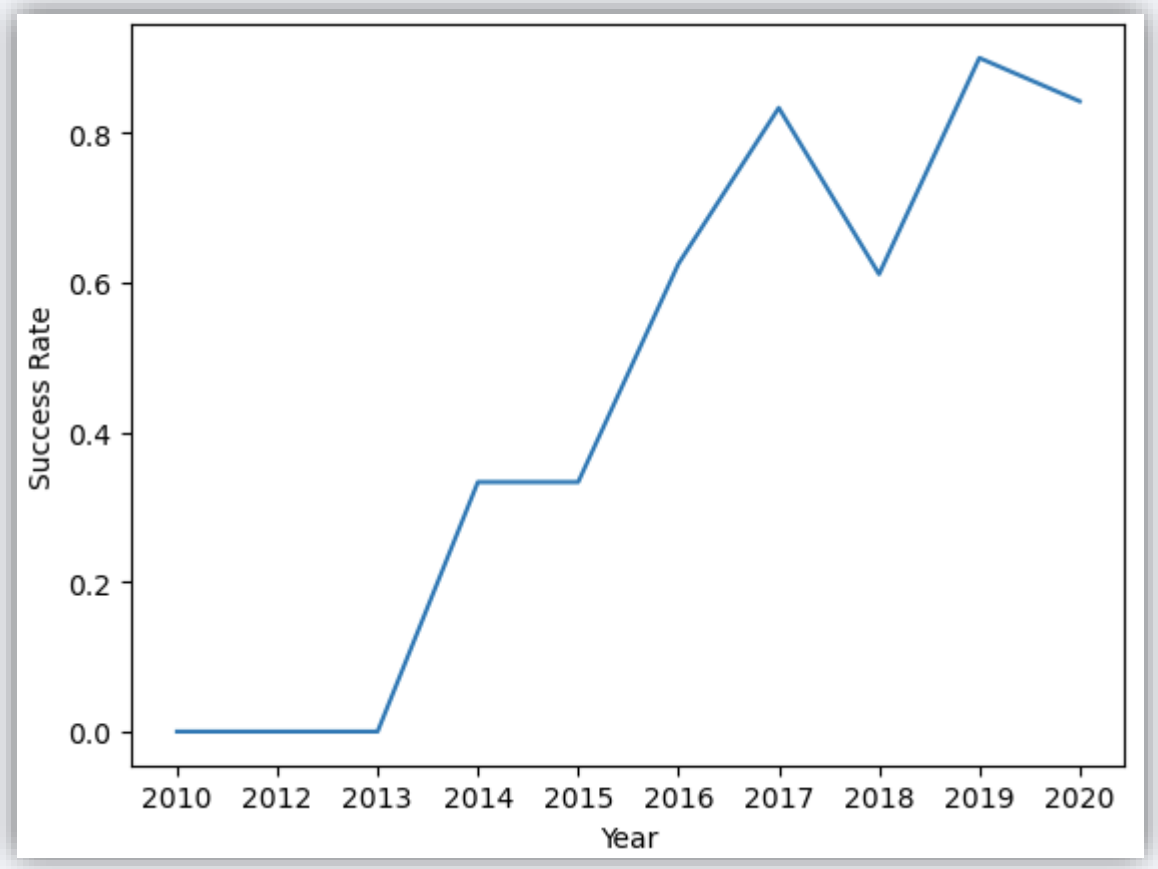
- The scatter plot shows the relationship between payload mass and orbit type
- Higher payloads have more successful landings in LEO, Polar and ISS orbits
- In GTO orbit, both success and failure occur, showing no clear pattern



Payload vs. Orbit Type scatter plot

Launch Success Yearly Trend

- The line chart shows the yearly launch success rate trend
- Success rate increases significantly after 2013
- Overall, success improves over time, indicating better landing technology and experience



Launch Success Yearly Trend Line Chart

All Launch Site Names

- This SQL query was used to find different launch sites, without duplicates.

%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

- The dataset contains four unique launch sites:
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- These sites represent different launch locations used by SpaceX for various missions

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Lauch Site Names

Launch Site Names Begin with 'CCA'

- The below SQL query was used to extract 5 records where launch site names begin with “CCA”

%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

- The results show launches from CCAFS LC-40, one of SpaceX’s primary launch sites
- This demonstrates how pattern filtering (LIKE ‘CCA%’) can be used to retrieve specific data

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, t
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	

Lauch Site Names that begin with ‘CCA’

Total Payload Mass

- The below SQL query was used to calculate the total payload mass launched by NASA (CRS)

%sql SELECT SUM(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';

- The total payload mass was found to be 45596 kg
- This demonstrates the use of the SUM() function to aggregate data

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(Payload_Mass__kg_)
45596

Total Payload Mass

Average Payload Mass by F9 v1.1

- This SQL query was used to calculate the average payload mass for Falcon 9 v1.1 boosters

%sql SELECT AVG(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';

- The average payload mass was approximately 2928.4 kg
- This was calculated using the AVG() function

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(Payload_Mass__kg_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(Payload_Mass__kg_)
2928.4

Average Payload Mass by F9 v1.1

First Successful Ground Landing Date

- This SQL query was used to find the first successful ground landing date

%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

- The first successful ground landing occurred on 22 December 2015 (2015-12-22)
- This was obtained using the MIN() function to get the earliest date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

First Successful Ground Landing Date

Successful Drone Ship Landing with Payload between 4000 and 6000

- This SQL query was used to find boosters that successfully landed on drone ships and the results were filtered for payload mass between 4000 kg and 6000 kg

```
%sql SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND Payload_Mass__kg_ > 4000 AND
Payload_Mass__kg_ < 6000;
```

- The query returned specific booster versions that meet these conditions using WHERE filters. The results were:
  - F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Boosters with Drone Ship Success (Payload 4000–6000)

Total Number of Successful and Failure Mission Outcomes

- The below SQL query was used to calculate the total number of mission outcomes grouped by landing outcome

```
%sql SELECT Landing_Outcome, COUNT(*) FROM SPACEXTABLE
GROUP BY Landing_Outcome;
```

- The results show:
  - Successful landings: 38
  - Success (drone ship): 14
  - Success (ground pad): 9
  - Failure: 3
  - Failure (drone ship): 5
  - Failure (parachute): 2
- This was calculated using COUNT() and GROUP BY

Landing_Outcome	COUNT(*)
Controlled (ocean)	5
Failure	3
Failure (drone ship)	5
Failure (parachute)	2
No attempt	21
No attempt	1
Precluded (drone ship)	1
Success	38
Success (drone ship)	14
Success (ground pad)	9
Uncontrolled (ocean)	2

Total Number of Successful and Failure Mission Outcomes

Boosters Carried Maximum Payload

- The below SQL query was used to identify boosters that carried the maximum payload mass

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Payload_Mass__kg_ =(SELECT MAX(Payload_Mass__kg_) FROM SPACEXTABLE);
```

- The results show that multiple boosters, mainly from the Falcon 9 Block 5 (F9 B5) series, achieved the maximum payload
- Results:

F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, F9 B5 B1049.7

- This was determined using the MAX() function with a subquery

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Boosters Carried Maximum Payload

2015 Launch Records

- The below SQL query was used.

```
%sql SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,1,4) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

- In 2015, there were 2 failed drone ship landing attempts
- These occurred in January and April
- Boosters involved: F9 v1.1 B1012 and B1015
- Both launches were from CCAFS LC-40

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

2015 Launch Records

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The below query was used

```
%sql SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC;
```

- The most frequent outcome was “No attempt” (10 cases)
- Drone ship landings show both success and failure (5 each)
- Ground pad successes were recorded 3 times
- Ocean landings (controlled/uncontrolled) and parachute failures were less frequent
- Overall, landing outcomes vary significantly, with early missions having more unsuccessful or no-attempt landings

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

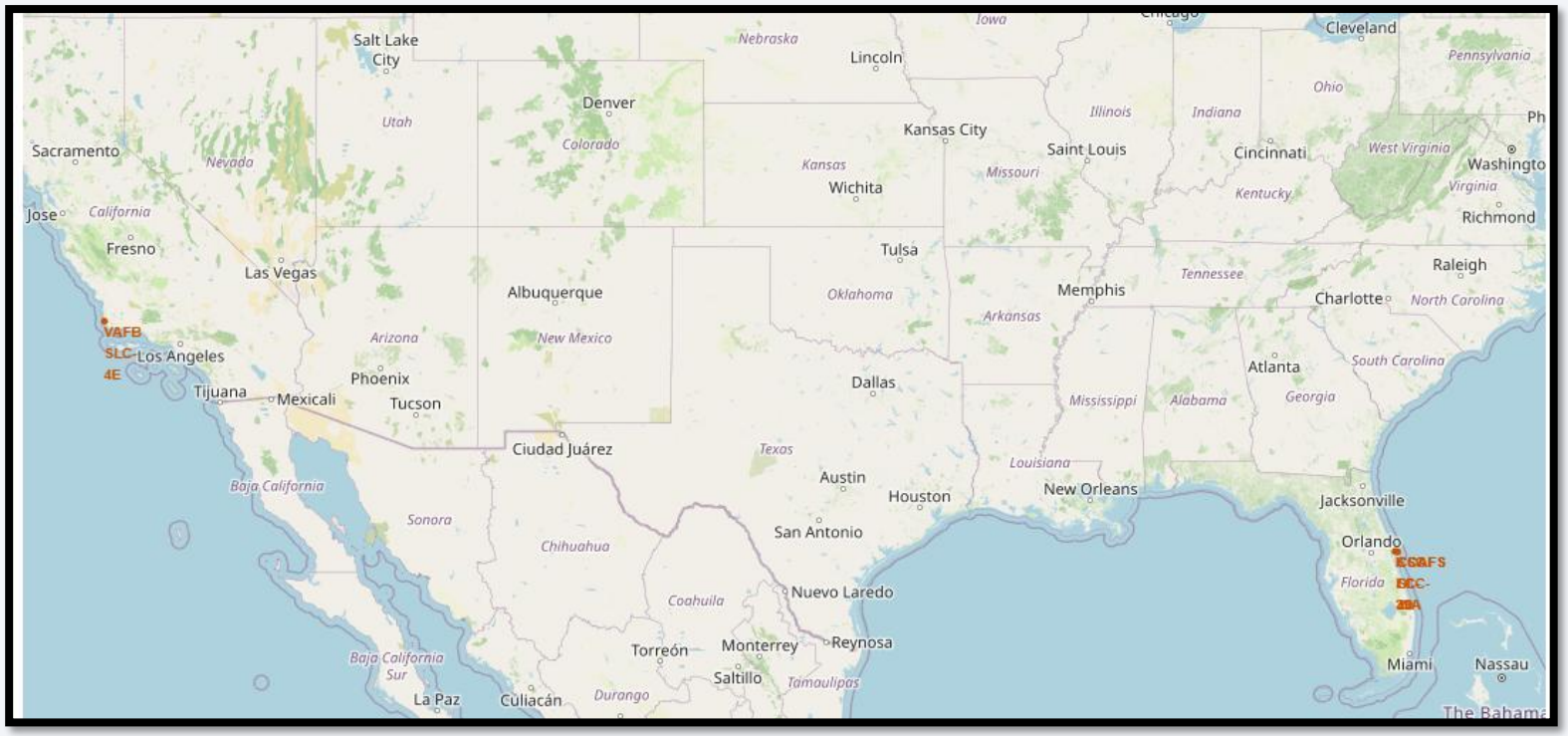
Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Section 3

Launch Sites Proximities Analysis

Launch Sites - Global Map Visualization

- The map shows the locations of SpaceX launch sites marked in orange.
- There are three main launch site regions visible:
  1. VAFB SLC-4E (California)
  2. CCAFS (Florida)
  3. KSC LC-39A (Florida)
- Two launch sites (CCAFS and KSC) are located very close to each other in Florida, forming a major launch hub. One launch site (VAFB) is located in California, on the west coast.



Launch Sites - Global Map Visualization

KEY FINDINGS

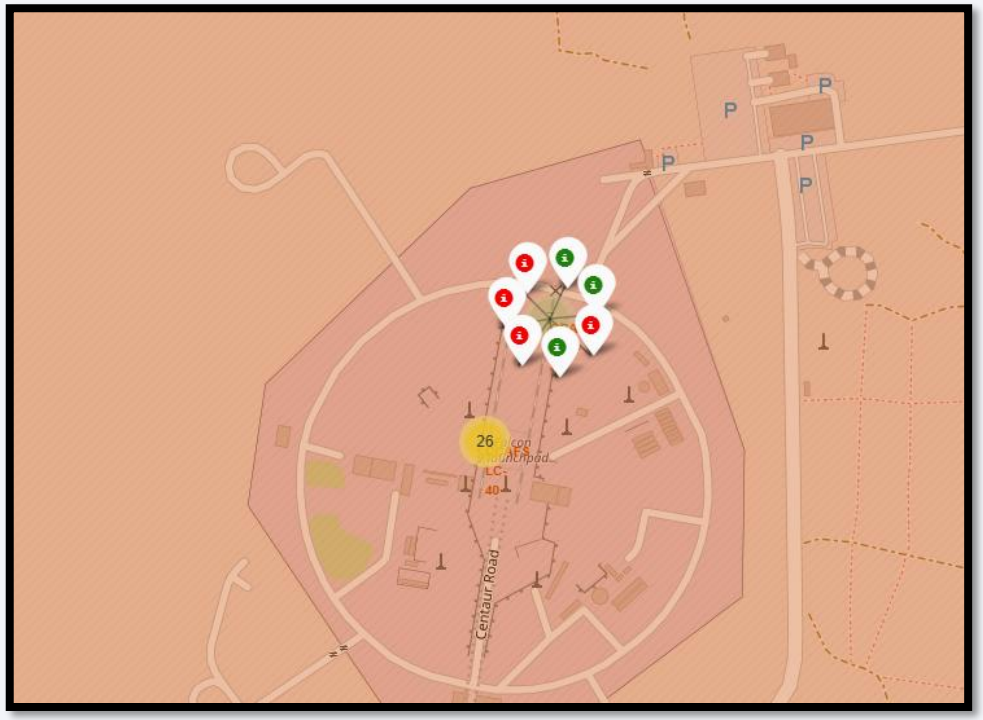
- All launch sites are located near coastlines, which ensures safety as rockets are launched over the ocean.
- Launch sites are not randomly distributed; they are strategically placed on the east and west coasts.
- These locations support efficient launches and reduced risk to populated areas.

Launch Outcomes at Selected Launch Site (CCAFS)

- The screenshot shows a zoomed-in view of a launch site with multiple launch pads.
- Each marker represents a launch, where:
  1. Green markers indicate successful launches
  2. Red markers indicate failed launches
- The markers are clustered around the launch complex, indicating repeated launch activity from the same site.

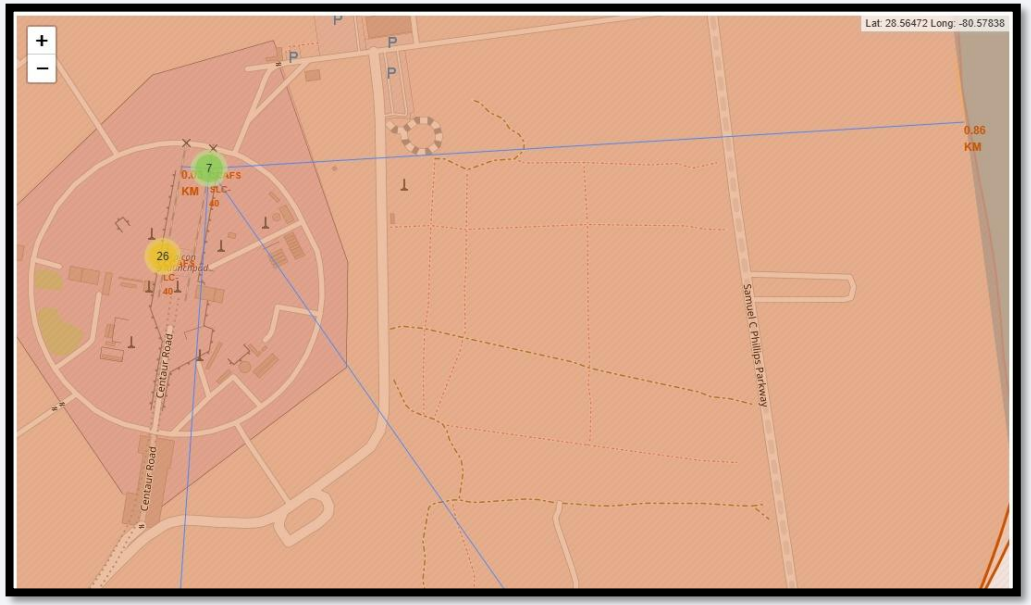
KEY FINDINGS

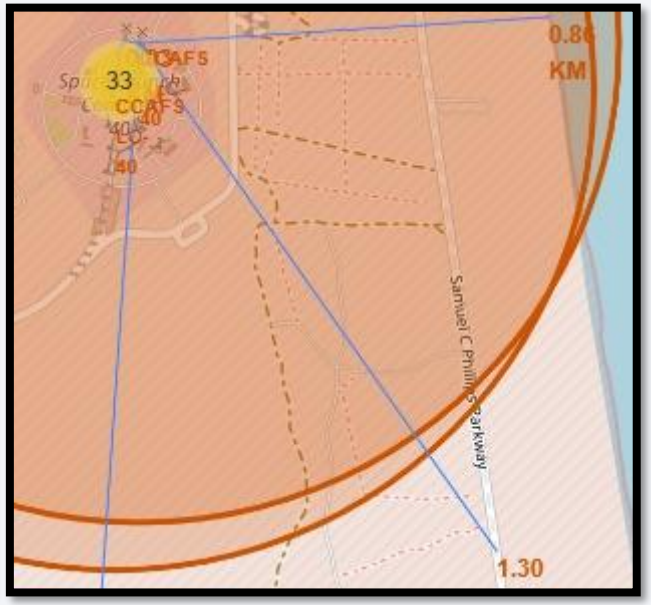
- In this screenshot, there are slightly more failed launches than successful launches at the visible site (4 red vs 3 green).
- This suggests that the selected site had a mixed performance, with failures slightly higher than successes.

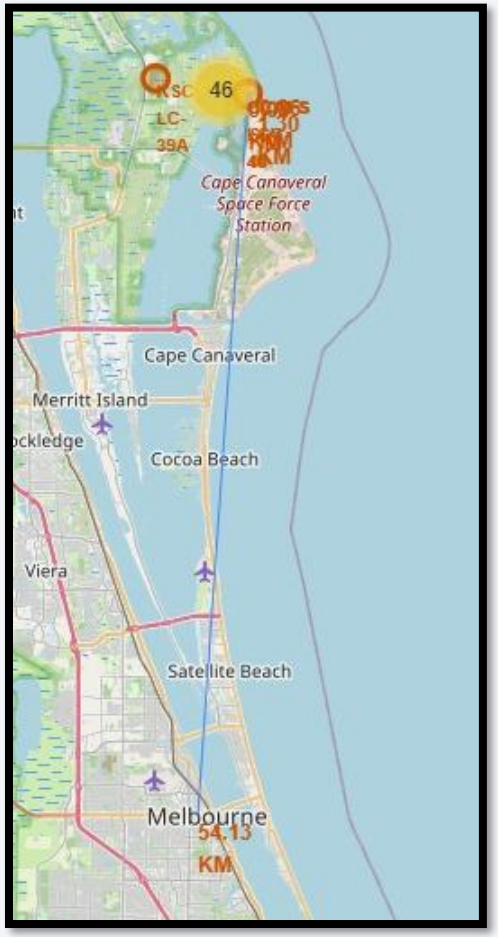


Launch Outcomes at Selected Launch Site (CCAFS)

Proximity Analysis of Launch Site (CCAFS SLC-40)







- The map shows a selected launch site (CCAFS SLC-40) and its distances to nearby features.
- The blue lines represent distances from the launch site to different proximities.

Proximity Analysis of Launch Site (CCAFS SLC-40)

KEY FINDINGS

- The coastline is very close to the launch site, at approximately 0.86 km.
- The highway is located at a distance of about 1.30 km from the launch site.
- The railway appears to be extremely close, with a distance close to 0.3 km .
- The map also highlights surrounding infrastructure such as roads and nearby areas.
- The site is far from the city (e.g., Melbourne - 54 km), which helps reduce risk to people in case of any failure.

CONCLUSION

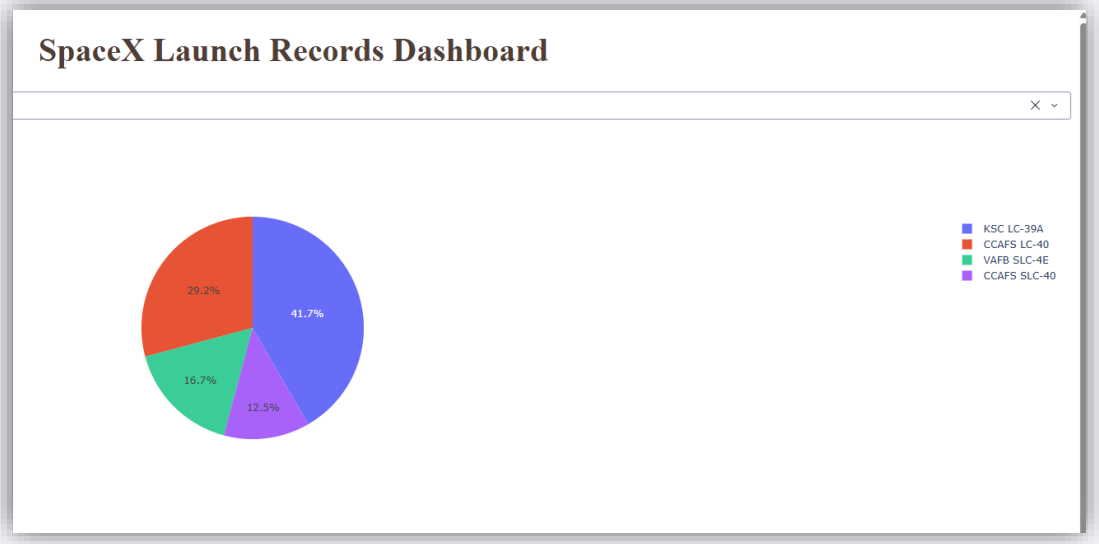
- This shows that the launch site is located very close to the coastline and transportation infrastructure.

Section 4

Build a Dashboard with Plotly Dash

Total Successful Launches by Site

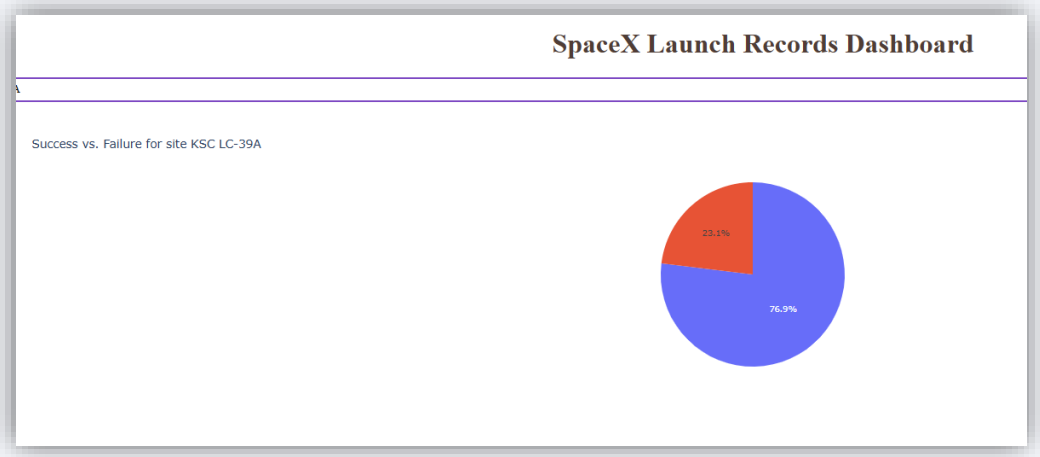
- The pie chart shows the distribution of successful launches across all launch sites.
- KSC LC-39A has the highest share of successful launches (about 41.7%), making it the top-performing site.
- CCAFS LC-40 is the second-highest contributor (around 29.2%).
- VAFB SLC-4E contributes a moderate portion (about 16.7%).
- CCAFS SLC-40 has the lowest share (around 12.5%).
- Overall, the chart highlights that launch success is not evenly distributed, with KSC LC-39A dominating in total successful launches



Total Successful Launches by Site

Launch Success Ratio - Highest Performing Launch Site

- The pie chart shows the success vs. failure ratio for the selected launch site (KSC LC-39A), which has the highest launch success rate among all sites.
- The chart indicates that approximately 76.9% of launches are successful, while about 23.1% are failures.
- This clearly shows that KSC LC-39A is the most reliable launch site, with a significantly higher success rate compared to failures.

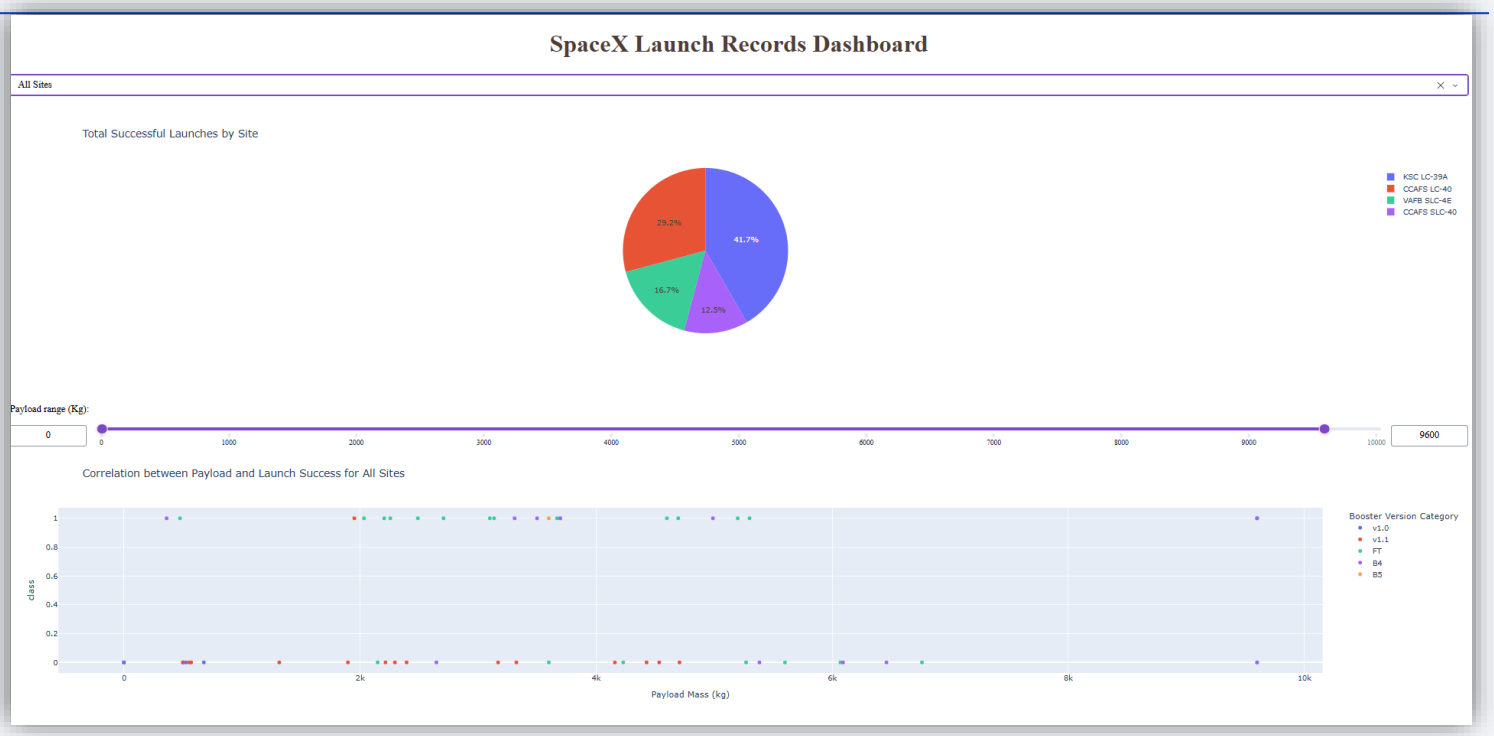


Launch Success Ratio - KSC LC-39A

- The dashboard also includes a payload range slider and a scatter plot, which help explore how payload mass may influence launch outcomes.
- Overall, the key finding is that KSC LC-39A has the highest success ratio, making it the best-performing launch site in the dataset.

Payload vs. Launch Outcome Analysis

- The scatter plot shows the relationship between payload mass (kg) and launch success (0 = failure, 1 = success) across all launch sites.
- Most successful launches occur between 2000 kg - 6000 kg
- Different booster versions (such as FT and B5) appear in successful launches



Payload vs. Launch Outcome Analysis

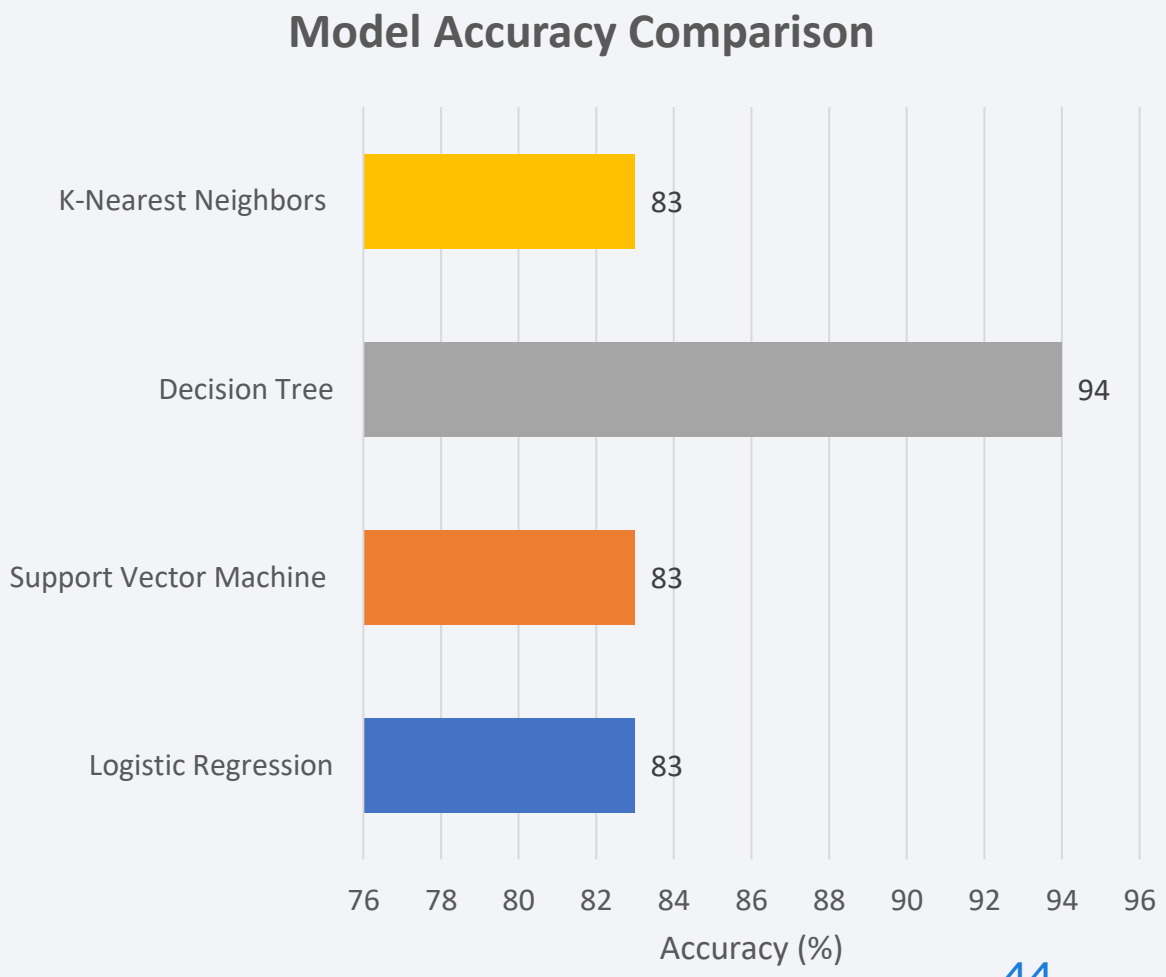
- Using the payload slider, we can observe that success patterns vary slightly across different payload intervals.
- There is no strong linear relationship between payload mass and launch success, as both successes and failures occur at different payload levels.

Section 5

Predictive Analysis (Classification)

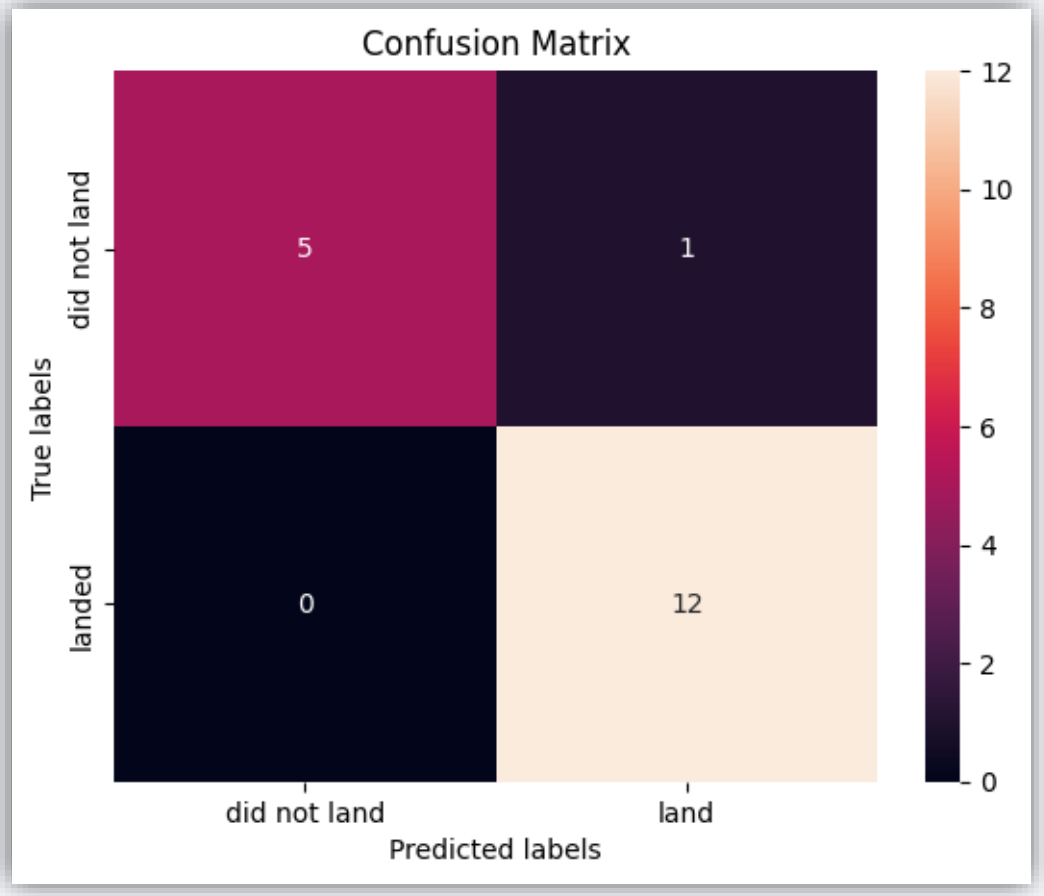
Classification Accuracy

- The accuracy of all classification models was visualized using a bar chart.
- The models compared include:
  1. Logistic Regression (~83%)
  2. Support Vector Machine (~83%)
  3. Decision Tree (~94%)
  4. K-Nearest Neighbors (~83%)
- The Decision Tree model achieved the highest classification accuracy among all models.



Confusion Matrix

- The confusion matrix shows the performance of the Decision Tree model, which is the best-performing model.
- The model correctly predicted:
  - 12 successful landings (True Positives)
  - 5 failed landings (True Negatives)
- The model made:
  - 1 false positive (predicted landing, but it did not land)
  - 0 false negatives (no missed successful landings)
- This indicates that the model performs very well in predicting successful landings, with very few errors.



Confusion matrix - Decision Tree model

Conclusions

- Machine learning models can effectively predict Falcon 9 landing outcomes.
- Data preprocessing and hyperparameter tuning improved model performance.
- The Decision Tree classifier achieved the highest accuracy (~94%).
- The confusion matrix confirms strong model performance with very few errors.
- This approach can help estimate launch success and reduce cost uncertainty.

Appendix

- Additional materials include model outputs, visualizations (bar chart, confusion matrix) and map screenshots used in the analysis.
- Python code and datasets were used from the provided lab environment.

Thank you!